

INPUTMIX: A STRATEGY TO REGULARIZE AND BALANCE MULTI-MODALITY AND MULTI-VIEW MODEL LEARNING

Junjie Wang^{*} Tomas Nordström^{*†}

^{*} Department of Applied Physics and Electronics, Umeå University, Sweden

[†]RISE Research Institutes of Sweden

ABSTRACT

Real-world perception tasks often involve multiple modalities or views of input. While joint training of multiple modality classification models has been explored previously, it has not consistently outperformed the best single modality model. This paper aims to address one of the reasons for this: the difficulty in balancing the contributions of each input in the end-to-end training of multi-input models. Additionally, the increased capacity of multi-input networks can lead to overfitting. To solve these issues, we propose InputMix, a simple yet effective method for optimally mixing different inputs. Our method mixes a certain proportion p of input pairs to relieve the increased capacity problems and assigns a weighting factor λ for each input to generate a mixed target, allowing us to specify the contributions of each input. Experimental results on three multi-input classification tasks demonstrate that our method significantly improves the generalization performance of multi-input neural networks. Codes are available at <https://github.com/JesseWong333/inputmix/>.

Index Terms— multi-view learning, multi-modality learning

1. INTRODUCTION

Deep neural networks have been shown to be highly effective models for solving a wide range of problems across multiple domains, including image, video, audio, and natural language [1, 2]. In real-world perception applications, it is common to collect data from multiple sensors to improve task performance within the same scene. The data can be multiple modalities (referred to as multi-modality) or uni-modalities captured from different viewpoints (referred to as multi-view). In this paper, we use the term *multi-input* to indicate both multi-modality and multi-view cases.

Existing research has pointed out that end-to-end training multi-input networks does not have to outperform the best single-input model [3]. We have also observed this phenomenon when training on SUN-RGBD dataset, scene classification task [4]. When jointly training with both RGB and Depth images, we can get an accuracy of 56.2%, which is similar to the RGB-only model (56.3%). This is against our intuition since the multi-input network receives additional information, it is expected to surpass the performance of its uni-modal counterpart. To investigate the non-intuitive results, explainable artificial intelligence (XAI) [5] method has been used. Specifically, we used the integrated gradients [6] method to estimate the contribution of each input. We began by computing an attribution map for each input, which identified the contribution of each pixel to the predicted label. Then, we used the sum of each attribution map as the contribution value of each input. We empirically found that RGB image contributes most of the predicted label, as shown in Fig. 1. On average, RGB images contributed 80% of the prediction

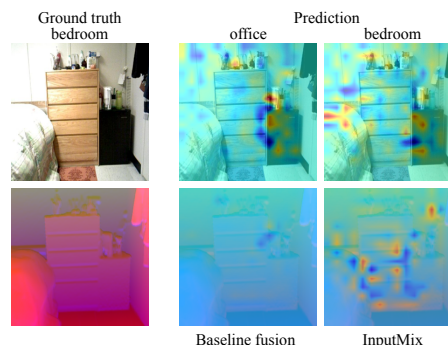


Fig. 1. Visualization of attribution maps. From the left to right column, we show the original RGB and depth images, the attribution maps of the baseline fusion method for its predicted class, and the attribution maps of InputMix, respectively. InputMix encourages the model to focus on both the RGB and depth modalities.

on the test set, while depth images only contributed 20%. The dominance of RGB image may also explain why RGB-only model and the joint training model have similar accuracy.

Another reason for the suboptimal performance lies in the overfitting issue. Sharing the parameters of feature extractors across multiple inputs usually limits the ability of representation. Traditionally, in multi-modal learning, we use separate feature extractors for different modalities to capture their unique characteristics. In this case, a late fusion RGB-Depth network has almost twice the parameters of an RGB-only network, and the increased number of parameters can lead to overfitting.

Existing research has tried to solve the problem from the perspective of overfitting. Gradient-Blending [3, 7] measures the change in overfitting and generalization for each branch, then proposes a training scheme to minimize the overfitting-to-generalization ratio. Regularization strategies for single input models, for example, early stopping, label smoothing [8] and mixup [9] could also alleviate the overfitting problem to some degree. On the other hand, one may speculate that the architectural deficiency for multi-input could be the cause. This can be solved either by hand-designed architecture [10, 11] or automatically searched architecture [12].

Based on our observation, we argue that an alternative way to solve the problem is to jointly balance the contribution of each input, and alleviate the overfitting issue. In this paper, we propose InputMix, a unified yet simple method that directly alleviates the two issues. In particular, we first mix the inputs and generate new training pairs that contain inputs belonging to different categories. By doing so, we actually augment training pairs but without manipulating the data of the original input. Therefore, we alleviate the overfitting. The corresponding targets of generated pairs change from one-hot vectors to soft targets. To balance different inputs, we

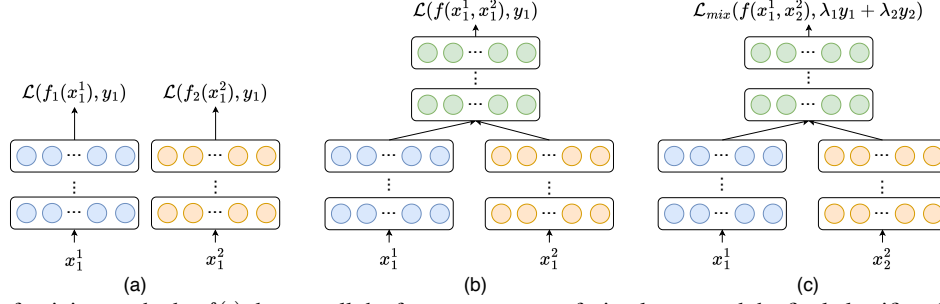


Fig. 2. Comparison of training methods. $f(\cdot)$ denotes all the feature extractors, fusion layers, and the final classifier. (a) Uni-modal training with two inputs x_1^1, x_1^2 . The losses are calculated separately. (b) joint training with two inputs from one paired sample $\{(x_1^1, x_1^2), y_1\}$. (c) joint training with one mixed pair $\{(x_1^1, x_2^2), \lambda_1 y_1 + \lambda_2 y_2\}$. InputMix alternates the training paradigms of (b) and (c).

then weigh the generated soft target proportionally. InputMix is a model-agnostic method that readily facilitates the training of diverse neural networks.

We used three multi-input datasets (AV-MNIST [13], SUN-RGBD [4], Compars [14]) to conduct the experiments. Various experiments show the effectiveness of the proposed methods. In summary, the contributions of this work include:

- We first use the XAI tool to identify one of the causes of sub-optimal performance in joint training of multi-input networks, which is that end-to-end training multi-input model could fail to balance the contributions of each input.
- We propose a unified yet simple method named InputMix that directly alleviates the unbalance and overfitting issues. We show that the proposed method significantly improves the generalization performance of multi-input neural networks on various datasets.

2. BACKGROUND

In general, given a training set $T = \{x_1^1, \dots, x_n^m, y_1, \dots, y_n\}$, where x_j^i is the i -th view of j -th training example, and y_j is the ground truth label, our goal is to train a model that using all the m inputs to produce a final prediction.

Uni-modal/view training. In the uni-modal/view training case, each branch does not fuse features. Training on one single input i is to minimize the following empirical loss:

$$\mathcal{L}(\mathcal{C}(\phi_i(x^i)), y) \quad (1)$$

where $\phi_i(\cdot)$ denotes the uni-modal/view feature extractor, $\mathcal{C}(\cdot)$ is the final classifier, and $\mathcal{L}$ is the loss function. Fig. 2a shows the uni-modal/view training with the two inputs of one sample $\{(x_1^1, x_1^2), y_1\}$ independently.

Joint training with fusion model. Consider CNN-based fusion neural networks with m view x^i , each input is processed by different feature extractors ϕ_m . The extracted features are fed to a fusion operation. For the CNN-based model, we use concatenation as the fusion operation. We stack fusion layers and a classifier to generate the final prediction. Formally, training the fusion neural network is to minimize the loss:

$$\mathcal{L}(\mathcal{C}(\phi_F([\phi_1(x^1), \phi_2(x^2), \dots, \phi_m(x^m)])), y) \quad (2)$$

where $\phi_F(\cdot)$ denotes fusion layers, $[\cdot]$ denotes the concatenation layer.

For the RGB and Depth modalities, we also adopt transformer-based fusion architectures, specifically utilizing the Vision Transformer (ViT) [15]. The ViT model extracts N non-overlapping

patches from the 2D image and converts them into a series of 1D tokens, as follows:

$$z^i = [\mathcal{E}^i(x^i), z_{cls}^i] + pos^i \quad (3)$$

where $\mathcal{E}$ is a linear projection layer that maps each patch to 1D token, and pos^i is a learned positional embedding added to the tokens to retain positional information. z_{cls} is a special token, and the final layer token z_{cls}^L can be passed to a classifier for classification tasks.

Considering a transformer-based fusion model with L layers, we define a starting fusion layer scalar L_f . When the current layer $l < L_f$, we do not exchange features between branches. For each branch i , we calculate the output of current transformer layer l as follows:

$$y^i(l) = \text{MHA}(z^i(l)) + z^i(l) \quad (4)$$

$$z^i(l+1) = \text{MLP}(y^i(l)) + y^i(l) \quad (5)$$

where MHA stands for Multi-Headed Attention as described in [16], and MLP stands for Multilayer Perceptron in the transformer model. When $l \geq L_f$, we start to fuse the features from different branches. Each token aggregates information from its own branch as well as from the other branch. We concatenate the outputs of different layers as follows:

$$z(l) = [z^1(l), z^2(l), \dots, z^m(l)] \quad (6)$$

Then, the output of the fusion layer is calculated using Eq. 4 and 5, except that we now have only one branch instead of m branches. The final loss function can be defined as:

$$\mathcal{L}(\mathcal{C}(z_{cls}^1(L) + z_{cls}^2(L), \dots, z_{cls}^m(L)), y) \quad (7)$$

Fig. 2b shows the joint training with one sample $\{(x_1^1, x_1^2), y_1\}$.

3. INPUTMIX FOR MULTI-INPUT MODELS

Let $\{x_j = (x_j^1, \dots, x_j^m), y_j\}$ be a training pair in the training set T , the core idea of InputMix is to generate new training pairs by optimally mixing multiple training pairs. Note that the mixing is done irrespective of target categories.

As an example, consider a scenario where there are two inputs, and we want to generate a mixed training sample. We start with two samples $\{(x_1^1, x_1^2), y_1\}$ and $\{(x_2^1, x_2^2), y_2\}$, and mix them to form a new training sample $\{(x_1^1, x_2^2), \lambda_1 y_1 + \lambda_2 y_2\}$ as shown in Fig. 2c.

To be more general, we use a permutation matrix $\pi^i \in R^{n \times n}$ to randomly permute each ordered set v^i containing all the i -th inputs of the n training pairs. We formulate a constraint condition to ensure the random permutation process always generates mixed training pairs instead of paired samples. The generated set of mixed training pairs can be obtained as follows:

$$\hat{\mathbf{X}} = [\pi^1(v^1), \dots, \pi^i(v^i)] \text{ s.t. } \forall e \in \sum_{i=1}^{i=n} \pi^i \neq m. \quad (8)$$

The generated pair $\hat{x}_j$ is the j -th row of $\hat{X}$.

For the generated pairs, we assign a weighting factor λ_i treated as a hyper-parameter to each input to generate a mixed target $\hat{y}_j$.

$$\hat{y}_j = \sum_{i=1}^{i=m} \lambda_i y_j^i \text{ s.t. } \sum_{i=1}^{i=m} \lambda_i = 1, \quad (9)$$

where y_j^i is the target of the i -th view of $\hat{x}_j$.

We only shuffle a certain proportion p of pairs, and for the rest of the pairs, we keep them unchanged. After randomly augmenting the data of each input, InputMix is applied within one mini-batch, similar to [9].

The differences with mixup [9]. Mixup is a technique for single-input models, which generates a new sample by linearly interpolating between two samples. The linear interpolations of the inputs and the associated labels share the same interpolations rate.

InputMix is dedicated to multi-input models. There is no parameter when mixing the inputs, while we do have a similar weighting parameter λ when generating labels. The purpose of the weighting parameter is to adjust the contribution of different inputs. In our experimental results, we show that adjusting the weighting parameter λ can affect the contribution and the classification performance. Furthermore, the InputMix technique is not limited to only two inputs. It's easy to scale up to many more inputs.

4. EXPERIMENTS

We experimented with three multi-input datasets, AV-MNIST [13], SUN-RGBD [4], and CompCars [14]. The datasets contain different multi-input scenarios including Grayscale image + Audio, RGB image + Depth image, and RGB image + RGB image, making them good choices for experimental testing in this field.

4.1. Datasets and experimental setups

AV-MNIST [13] is a multimedia dataset assembled from images of handwritten digits [17] and audio samples of spoken digits [18]. The prediction task is identical to the MNIST dataset, with the goal of predicting each digit into one of ten classes. The image modality was altered by removing 75% of the energy using Principal Component Analysis (PCA), and the audio modality was augmented with natural noise. The final generated dataset has the same number of examples as MNIST with 55,000 training, 5,000 validation, and 10,000 testing examples. We used LeNet-3 as the image encoder, LeNet-5 as the audio encoder, and concatenation to fuse features from the image encoder and audio encoder, as used in [2].

SUN-RGBD [4] dataset is a scene understanding dataset consisting of RGB-D images from NYU depth v2 [19], Berkeley B3DO [20], and SUN3D [21]. The dataset includes a total of 10,335 RGB-D images. Following the common practice [22], we limited our focus to the 19 primary scene categories and obtained a total of 4,845 training samples and 4,659 test samples.

We utilized a ResNet-18 [23] backbone for the CNN-based fusion model and stacked 2 linear layers to process the fused features further and generate final outputs. The backbone architecture of the transformer-based fusion model follows ViT-S [15], initialized with weights pre-trained on imagenet1k [24]. Depth images employed two preprocessing methods: HHA [25] that encodes depth with 3 channels, and a scaling method that directly scales grayscale depth images to the 0 to 255 range. All modalities are pre-processed with standard data augmentations, including random crop and flip.

CompCars [14] comprises web-nature and surveillance-nature images of cars, with web-nature images being annotated with car maker, car model, year, and viewpoint, where each viewpoint can be

Table 1. Digital classification tasks on AV-MNIST dataset. All experiments were repeated 10 times and a mean and standard deviation were reported. We compare the methods with the same neural architecture.

Methods	Validation set	Test set
Unimodal(Image)	69.59±0.23	65.1±0.2
Unimodal(Audio)	42.22±0.03	42.0±0.2
Late Fusion	74.63±0.19	71.7±0.4
LRTF[26]	74.67±0.26	71.5±0.5
MI-MATRIX[27]	74.13±0.30	71.2±0.5
CCA[28]	75.01±0.19	71.9±0.4
ours	76.02±0.17	72.0±0.3

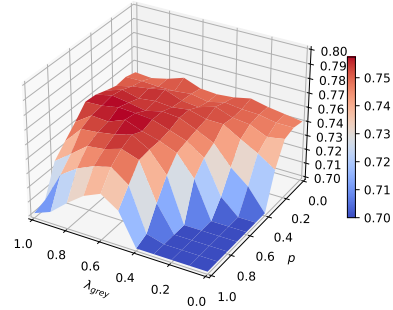


Fig. 3. The influence of the hyper-parameters p and λ_{grey} on the AV-MNIST dataset. Both the two hyper-parameters are set between [0.0, 1.0] with a step of 0.1. For better contrast, when the accuracy is below 0.7, we manually set it to 0.7

front, front-side, rear, rear-side, or side. To create multi-view pairs, we selected one image per viewpoint from the web-nature images. We chose 1,010 car models, without considering the year, and each category had a minimum of 5 samples, resulting in a total of 11,880 pairs. We split them into 7,856 training pairs and 4,024 test pairs. We used the same transformer-based fusion model as used for the SUN-RGBD dataset.

4.2. Results

AV-MNIST. Table 1 shows the performance on AV-MNIST dataset. The re-implementations of the compared methods are based on [2]. By setting the hyper-parameter $p=0.6$ and $\lambda_{grey}=0.7$ (weighting factor assigned to grey images), the model trained with InputMix is 1 point better than the compared method on the validation set and also shows improvement on the test set. Note that our method doesn't require any additional parameters or calculations, which have the exact same inference cost as the late fusion method.

Fig. 3 shows the influence of the hyper-parameters p and λ_{grey} on the AV-MNIST dataset. We see that the figure is convex and there exists an optimal combination of the two hyper-parameters. In this case, it's (0.6, 0.7). When setting $p=0.$, the case is equal to the base fusion method, and the validation accuracy is 74.63±0.19. This result indicates that the base fusion is a suboptimal solution compared to the optimal combination of p and λ . On the other hand, setting $p=1$ results in significantly lower performance. In that case, all the training data are mixed, while the validating data are not mixed. We believe the data discrepancy is the cause of the poor performance.

SUN RGB-D. We compared the proposed model with the existing methods on SUN RGB-D dataset, scene classification tasks. The experimental results are shown in table 2. Both single-input and

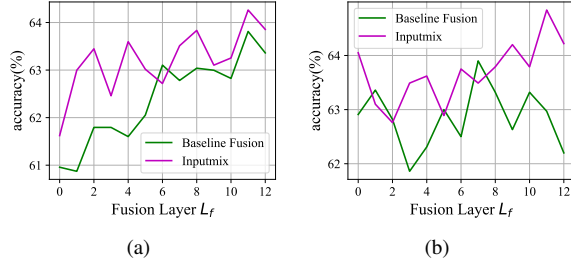


Fig. 4. The accuracy of InputMix and baseline fusion when fusing at various layers within the ViT-S based (12 layers) fusion model, experimented on SUN RGB-D dataset. (a) RGB+HHA. (b) RGB+grey.

fusion results are reported. The proposed method with ResNet-18 backbone achieves an accuracy of 61.5%, which matches the result of [22]. Compared with ResNet-18 based fusion network, the accuracy with ViT-S backbone increases by 3.2 percent points. The performance of both RGB and depth modality outperforms the ResNet-18 based method, which indicates that a more powerful backbone model still facilitates scene classification tasks. Table 3 shows the result of different training settings. From the table, we see that InputMix constantly achieves better performance than the base fusion method. Compared to the transformer-based fusion model, the CNN-based model has a more serious un-balanced issue. For the CNN-based model, baseline fusion achieves worse or similar performance to the RGB-only model, and InputMix could improve the baseline fusion by a large margin. While the transformer-based baseline fusion model is already 3 percent point better than the best single input model. Inputmix could still achieve better performance than baseline fusion.

Table 2. Comparison with current methods on SUN RGB-D dataset, scene classification tasks.

Methods	RGB	Depth	Fusion
Du et al.[29]	42.6	43.3	53.3
Li et al.[30]	46.3	39.2	54.6
Song et al.[31]	44.6	42.7	53.8
Ayub et al.[32]	48.8	37.3	59.5
Han et al.[22]	56.2	52.7	61.3
ours(ResNet-18 based, as in [22])	56.3	53.1	61.5
ours(ViT-S based)	61.4	54.2	64.3

Table 3. Comparison of training methods on SUN RGB-D dataset, scene classification tasks. The baseline is the fusion model trained without InputMix. HHA and grey indicate the processing method for depth images.

Backbone	RGB	Depth	Baseline	InputMix
ResNet-18 (HHA)	56.3	53.1	56.2	61.3
Vit-S (HHA)	61.4	54.2	63.9	64.3
ResNet-18 (grey)	56.3	48.9	55.4	60.5
Vit-S (grey)	61.4	52.5	63.9	64.8

Fig. 4 shows the benefits of InputMix over the baseline fusion method when fusing at various levels. In the two settings, the proposed method achieves better performance than the base fusion method for most of the fusion layers. The results show that the proposed method also works for different fusion levels.

To verify the hypothesis that changing hyper-parameters λ could guide to the contribution of each view, we did the following experiments. We first fixed the hyper-parameters $p=0.5$, and then changed

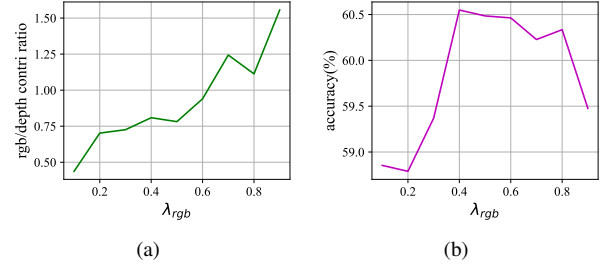


Fig. 5. The impact of hyper-parameters λ on the contribution ratio (a) and classification accuracy (b), experimented with CNN-based fusion model on SUN RGB-D dataset.

λ_{rgb} from 0.1 to 0.9, as shown in Fig. 5. As λ_{rgb} increases, the average RGB/depth contribution ratio on the test set also increases. When RGB or Depth dominates the contribution, the classification accuracy drops dramatically.

CompCars. We then investigated the scalability of InputMix. When adding more views to the fusion model, we always chose the worst view, as shown in the single view classification results, as seen in table 4. For example, for the 2 views model, we used the side and front-side views. For the 3 view model, we used the side, front-side, and rear-side views. From Fig. 6, we see that the classification accuracy increases with the number of views. The InputMix method outperforms the baseline fusion model by 3-6 percent points when fusing 2-5 views. The results demonstrate that our method is effective in multi-view scenarios and can also be scaled up to more views.

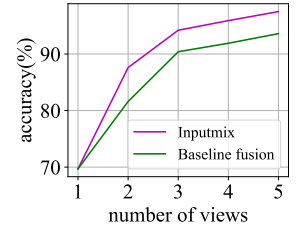


Fig. 6. The classification accuracy on CompCars dataset when scaled to 5 views.

Table 4. Single view classification accuracy on CompCars dataset

View	side	front-side	rear-side	front	rear
accuracy(%)	69.7	70.2	81.0	82.2	88.2

5. CONCLUSION

In conclusion, this paper proposes a unified and simple method called InputMix to address two main issues in multi-input classification networks, namely unbalanced contributions of multi-input and overfitting. The proposed method mixes input pairs to relieve the capacity problems and generates a soft target weighted proportionally to specify the contributions of each input. Experiment results on three multi-input classification tasks demonstrate that InputMix significantly improves the generalization performance of multi-input neural networks. The proposed method's simplicity and effectiveness make it a promising approach for real-world perception involving multi-view or multi-modality inputs.

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6. REFERENCES

- [1] Arnab Barua, Mobyen Uddin Ahmed, and Shahina Begum, “A systematic literature review on multimodal machine learning: Applications, challenges, gaps and future directions,” *IEEE Access*, vol. 11, pp. 14804–14831, 2023.
- [2] Paul Pu Liang, Yiwei Lyu, Xiang Fan, Zetian Wu, Yun Cheng, Jason Wu, Chen, et al., “Multibench: Multiscale benchmarks for multimodal representation learning,” in *Proc. NeurIPS Track on Datasets and Benchmarks*, 2021.
- [3] Weiyao Wang, Du Tran, and Matt Feiszli, “What makes training multi-modal classification networks hard?,” in *Proc. CVPR*, 2020, pp. 12695–12705.
- [4] Shuran Song, Samuel P Lichtenberg, and Jianxiong Xiao, “Sun rgb-d: A rgb-d scene understanding benchmark suite,” in *Proc. CVPR*, 2015, pp. 567–576.
- [5] Rudresh Dwivedi, Devam Dave, Het Naik, Smiti Singhal, Rana Omer, Pankesh Patel, Bin Qian, Wen, et al., “Explainable ai (xai): Core ideas, techniques, and solutions,” *ACM Comput. Surv.*, vol. 55, no. 9, pp. 1–33, 2023.
- [6] Mukund Sundararajan, Ankur Taly, and Qiqi Yan, “Axiomatic attribution for deep networks,” in *Proc. ICML*, 2017, pp. 3319–3328.
- [7] Huy Phan, Huy Le Nguyen, Oliver Y Chén, Lam Pham, Philipp Koch, Ian McLoughlin, and Alfred Mertins, “Multi-view audio and music classification,” in *Proc. ICASSP*, 2021, pp. 611–615.
- [8] Rafael Müller, Simon Kornblith, and Geoffrey E Hinton, “When does label smoothing help?,” in *Proc. NeurIPS*, 2019, pp. 4696–4705.
- [9] Hongyi Zhang, Moustapha Cissé, Yann N. Dauphin, and David Lopez-Paz, “mixup: Beyond empirical risk minimization,” in *Proc. ICLR*, 2018.
- [10] Arsha Nagrai, Shan Yang, Anurag Arnab, Aren Jansen, Cordelia Schmid, and Chen Sun, “Attention bottlenecks for multimodal fusion,” in *Proc. NeurIPS*, 2021, vol. 34, pp. 14200–14213.
- [11] Jiaming Zhang, Huayao Liu, Kailun Yang, Xinxin Hu, Ruiping Liu, and Rainer Stiefelwagen, “Cmx: Cross-modal fusion for rgb-x semantic segmentation with transformers,” *IEEE Trans. Intell. Transp. Syst.*, pp. 1–16, 2023.
- [12] Juan-Manuel Pérez-Rúa, Valentin Vielzeuf, Stéphane Pateux, Moez Baccouche, and Frédéric Jurie, “Mfas: Multimodal fusion architecture search,” in *Proc. CVPR*, 2019, pp. 6966–6975.
- [13] Valentin Vielzeuf, Alexis Lechervy, Stéphane Pateux, and Frédéric Jurie, “Centralnet: a multilayer approach for multimodal fusion,” in *Proc. ECCV Workshops*, 2018, pp. 575–589.
- [14] Linjie Yang, Ping Luo, Chen Change Loy, and Xiaoou Tang, “A large-scale car dataset for fine-grained categorization and verification,” in *Proc. CVPR*, 2015, pp. 3973–3981.
- [15] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, et al., “An image is worth 16x16 words: Transformers for image recognition at scale,” in *Proc. ICLR*, 2021.
- [16] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Lukasz Kaiser, and Illia Polosukhin, “Attention is all you need,” in *Proc. NeurIPS*, 2017, pp. 5998–6008.
- [17] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner, “Gradient-based learning applied to document recognition,” in *Proc. IEEE*, 1998, vol. 86, pp. 2278–2324.
- [18] R Gary Leonard and George Doddington, “Tidigits speech corpus,” *Texas Instruments, Inc*, 1993.
- [19] Nathan Silberman, Derek Hoiem, Pushmeet Kohli, and Rob Fergus, “Indoor segmentation and support inference from rgb-d images,” in *Proc. ECCV*, 2012, pp. 746–760.
- [20] Allison Janoch, Sergey Karayev, Yangqing Jia, Jonathan T Barron, Mario Fritz, Kate Saenko, and Trevor Darrell, “A category-level 3d object dataset: Putting the kinect to work,” in *Consumer depth cameras for computer vision*, pp. 141–165. Springer, 2013.
- [21] Jianxiong Xiao, Andrew Owens, and Antonio Torralba, “Sun3d: A database of big spaces reconstructed using sfm and object labels,” in *Proc. ICCV*, 2013, pp. 1625–1632.
- [22] Zongbo Han, Changqing Zhang, Huazhu Fu, and Joey Tianyi Zhou, “Trusted multi-view classification with dynamic evidential fusion,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 2551–2566, 2023.
- [23] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun, “Deep residual learning for image recognition,” in *Proc. CVPR*, 2016, pp. 770–778.
- [24] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin, “Emerging properties in self-supervised vision transformers,” in *Proc. ICCV*, 2021, pp. 9650–9660.
- [25] Saurabh Gupta, Ross Girshick, Pablo Arbeláez, and Jitendra Malik, “Learning rich features from rgb-d images for object detection and segmentation,” in *Proc. ECCV*. Springer, 2014, pp. 345–360.
- [26] Zhun Liu, Ying Shen, Varun Bharadhwaj Lakshminarasimhan, Paul Pu Liang, Amir Zadeh, and Louis-Philippe Morency, “Efficient low-rank multimodal fusion with modality-specific factors,” in *Proc. ACL*, 2018, pp. 2247–2256.
- [27] Siddhant M. Jayakumar, Wojciech M. Czarnecki, Jacob Menick, Jonathan Schwarz, et al., “Multiplicative interactions and where to find them,” in *Proc. ICLR*, 2020.
- [28] Zhongkai Sun, Prathusha Sarma, William Sethares, and Yingyu Liang, “Learning relationships between text, audio, and video via deep canonical correlation for multimodal language analysis,” in *Pro. AAAI*, 2020, vol. 34, pp. 8992–8999.
- [29] Dapeng Du, Xiangyang Xu, Tongwei Ren, and Gangshan Wu, “Depth images could tell us more: Enhancing depth discriminability for rgb-d scene recognition,” in *Proc. ICME*, 2018, pp. 1–6.
- [30] Yabei Li, Junge Zhang, Yanhua Cheng, Kaiqi Huang, and Tieniu Tan, “Df 2 net: Discriminative feature learning and fusion network for rgb-d indoor scene classification,” in *Proc. AAAI*, 2018, pp. 7041–7048.
- [31] Xinhang Song, Shuqiang Jiang, Luis Herranz, and Chengpeng Chen, “Learning effective rgb-d representations for scene recognition,” *IEEE Trans Image Process*, vol. 28, no. 2, pp. 980–993, 2018.
- [32] Ali Ayub and Alan R. Wagner, “Centroid based concept learning for RGB-D indoor scene classification,” in *Proc. BMVC*, 2020.